

Development Roadmap

Stage-I Completed

- ▶ IR retrieval / reranking
- ▶ Coord-aware encoding
- ▶ Distance-bias attention
- ▶ Forward-spectrum check
- ▶ 200 NIST spectra validated



Stage-II Planned

- ▶ Graph diffusion generator
- ▶ Size-adaptive generation
- ▶ Pairwise distance matrix
- ▶ Classical MDS 2D->3D



Stage-III Future

- ▶ TFN / SE(3) refinement
- ▶ Parity channels (0e,0o,1e,1o)
- ▶ IR / Raman / NMR / MS
- ▶ Top-K with phys. consistency